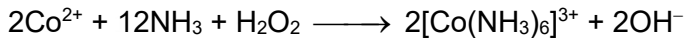


- 11** The equation for the formation of hexaamminecobalt(III), $[\text{Co}(\text{NH}_3)_6]^{3+}$ complex ion is shown below.



The rate equation is $\text{rate} = k [\text{Co}^{2+}][\text{NH}_3]^3[\text{H}_2\text{O}_2]$.

When the concentration of each reactant is $x \text{ mol dm}^{-3}$, the initial rate was found to be $y \text{ mol dm}^{-3} \text{ s}^{-1}$.

What will be the initial rate of the reaction if $[\text{Co}^{2+}]$ is $2x$, $[\text{NH}_3]$ is $\frac{1}{2}x$ and $[\text{H}_2\text{O}_2]$ is $2x$?

A $\frac{1}{8}y$

B $\frac{1}{2}y$

C $2y$

D $8y$